

Google new Research Paper : Measuring the environmental impact of delivering AI

Posted by u/Technical-Love-8479 • 0 points • 27 comments

Google has dropped in a very important research paper measuring the impact of AI on the environment, suggesting how much carbon emission, water, and energy consumption is done for running a prompt on Gemini. Surprisingly, the numbers have been quite low compared to the previously reported numbers by other studies, suggesting that the evaluation framework is flawed.

Google measured the environmental impact of **a single Gemini prompt** and here's what they found:

0.24 Wh of energy

0.03 grams of CO₂

0.26 mL of water

Paper :

[https://services.google.com/fh/files/misc/measuring_the_environmental_impact_of_delivering_ai_at_google_scale.pdf]

(https://services.google.com/fh/files/misc/measuring_the_environmental_impact_of_delivering_ai_at_google_scale.pdf)

Video : <https://www.youtube.com/watch?v=q07kf-UmjQo>

Comments

u/kaggleqrdl • 1 points • 2025-08-31 21:21 UTC

So, 3.6T tokens last week on openrouter. Assume an average of 1000 tokens per request, that is 936K litres of water per week. So basically a small lake every month or so. And that's just openrouter!

u/BusRevolutionary9893 • 1 points • 2025-08-31 15:01 UTC



u/lolzinventor • 1 points • 2025-08-24 17:53 UTC

[https://preview.redd.it/jdkynca3a0lf1.png?](https://preview.redd.it/jdkynca3a0lf1.png?width=1940&format=png&auto=webp&s=32c4d7a378ec5940e0b097d64537888a7cc0700e)

[width=1940&format=png&auto=webp&s=32c4d7a378ec5940e0b097d64537888a7cc0700e](https://preview.redd.it/jdkynca3a0lf1.png?width=1940&format=png&auto=webp&s=32c4d7a378ec5940e0b097d64537888a7cc0700e)

u/bick_nyers • 1 points • 2025-08-24 14:52 UTC

It takes about 0.5-2.5 kWh to manufacture a kg of plastic so I'd say this is pretty reasonable as far as footprint goes for the typical person.

Desalination is about 10 watts per gallon as well.

u/MuiaKi • 1 points • 2025-08-24 12:52 UTC

How many tokens per prompt?

I think local is still going to be more environmentally friendly in the long run since you don't waste drinking water in cooling your pc, much less your phone...

Solar + storage makes more sense at the individual level since you won't be subsidizing data centers.

And no one's installing generators just to run your devices.

u/whichkey45 • 2 points • 2025-08-24 08:10 UTC

Hmm Google tell us their AI is much less environmentally harmful than every other study!!

u/No_Efficiency_1144 • 1 points • 2025-08-24 06:59 UTC

When a Google framework gets different results I would at least give them the benefit of the doubt long enough to check their numbers.

u/Sufficient-Past-9722 • 4 points • 2025-08-24 06:31 UTC

Let's keep in mind that Google's data centers are easily the most efficient in the industry, so others are likely producing more CO2.

u/spacebrry • 1 points • 2025-08-24 15:16 UTC

source?

u/Sufficient-Past-9722 • 2 points • 2025-08-24 15:25 UTC

Anecdotal, I used to work there and saw how every last component in a data center is custom-planned, without needing to account for, especially, heterogeneous airflow patterns between rows and unpredictable/uncontrollable electric load from colo customers. The only other DC operators that can come close are Meta, Amazon, and Microsoft, with everyone else using primarily commodity cooling and power distribution.

u/Lissanro • 6 points • 2025-08-24 06:09 UTC

I run Kimi K2 locally (the 1T model, IQ4 quant with ikllama.cpp) on EPYC 7763 with 1TB RAM + 4x3090 (96GB VRAM). In my case, a typical result would be around 1K tokens - it is 38 Wh per every thousand of tokens.

It is a bit unclear though what exact length of a single prompt the paper compares against, it is not mentioned in the post, maybe I missed it in the paper, but searching it for "length" or "token" does not seem to find much. Also, I cannot find total and active parameter count of the model in question. So hard to say how it compares. Or maybe I missed something.

This is the only clue I found about typical response length to a prompt, about Mistral Large 2 model:

>Mistral AI, 2025 [18]: A peer-reviewed lifecycle assessment (LCA) for its Mistral Large 2 model was conducted in collaboration with the French environmental agency ADEME and consulting firm Carbone 4. For a typical 400-token response from its "Le Chat" assistant, Mistral reports a marginal impact of 1.14 grams of CO2e, and 45 milliliters (mL) of water consumed

It does not say how much Wh consumed, but given much larger "water" and "CO2" figures compared to "Gemini prompt", probably at least few Wh... which seems quite a lot for a cloud API where everything should be highly parallelized and batch processed.

Compared to my rig, which can run Mistral Large 123B 5bpw at 36-42 tokens/s (assuming tensor parallelism and speculative decoding enabled), I spend around 3 Wh per 400 token response, so it sounds like "Le Chat" at the time when it was measured wasn't running very efficient infrastructure, or maybe I am miscalculating something, since I would expect cloud API server to consume less than Wh for generating 400 tokens with 123B model.

If someone can point out what I missed or misunderstood, and share more exact numbers to compare against, I would be very interested to hear that!